

Wellhead Equipment Data Sheet—General						
Well name(s) and location(s): _____						
Maximum operating pressure: _____						
Anticipated wellhead shut-in pressure: _____						
Temperature ranges anticipated: _____						
Minimum ambient temperature: _____						
Maximum flowing fluid temperature at wellhead: _____						
Anticipated composition of produced fluids: CO ₂ _____ (mg) _____ Chlorides _____ (mg)						
_____ H ₂ S _____ (mg) _____ Other _____						
Water or brine pH: _____						
Does NACE MR0175/ISO 15156 apply? _____						
Anticipated production rates: _____ m ³ /d oil/condensate						
_____ m ³ /d gas						
_____ m ³ /d S&W ^a						
Will erosion be a concern? Cause: _____						
Will scale, paraffin, corrosion, or other types of inhibitors be used? _____						
Inhibitor type: _____ Inhibitor carrier: _____ Batch or continuous inhibition? _____						
Will acidification be performed? _____ Type of acid: _____						
External coating? Yes, type _____ No _____						
Internal coating? Yes, type _____ No _____						
Delivery requirements: _____						
Special shipping, packing, and storage instructions: _____						
Casing program						
		Top Joint in String				
	Size (OD)	kg/m (lb/ft)	Grade	Connection	Max./Min. Casing Load daN (lb)	Bit Size mm (in.)
Conductor	_____	_____	_____	_____	_____	_____
Surface casing	_____	_____	_____	_____	_____	_____
Protective casing	_____	_____	_____	_____	_____	_____
Production casing	_____	_____	_____	_____	_____	_____
Tubing	_____	_____	_____	_____	_____	_____
Type of completion: single or multiple	_____	_____	_____	_____	_____	_____
FOOTNOTE						
^a Sand and water.						

Figure B.1—Wellhead Equipment Data Sheet—General

Wellhead Equipment Data Sheet—Casing-head Housing			
Casing-head housing	PSL: _____	PR: _____	
Bottom connector:	Size: _____		
	Rated working pressure: _____		
	Type: _____		
Top connector:	Size: _____		
	Rated working pressure: _____		
	Type: _____		
Outlets:	Size: _____		
	Rated working pressure: _____		
	Type: _____		
	Number: _____		
Equipment for outlets:	Valve-removal plug: _____		
	Valves (right): Qty _____ Typ _____ PSL _____ PR _____		
	Valves (left): Qty _____ Typ _____ PSL _____ PR _____		
	Companion flanges: Qty _____ Typ _____ PSL _____		
	Bullplugs: Qty _____		
	Nipples: Qty _____		
	Needle valves: Qty _____		
	Gauges: Qty _____		
Lock screws? Yes _____ No _____	Lock screw function: _____		
Baseplate requirements: _____			
Special material requirements: _____			
Casing hanger: Size:	Type: _____		
	PSL: _____		
	PR: _____		
	Temperature rating (Table 2): _____		
Material class (Table 3): _____			
Retained fluid corrosivity (Table B.1): _____			
Witness? Yes ^a _____		No _____	
External coating? No _____ Yes _____	If yes, type _____		
Internal coating? No _____ Yes _____	If yes, type _____		
Flange bolting requirements (Table 16)	Nonexposed _____	Exposed _____	Exposed (low strength) _____
Main run (studs): _____	(nuts): _____		
Outlet inboard (studs): _____	(nuts): _____		
Outlet other (studs): _____	(nuts): _____		
Test and auxiliary equipment:			
Wear bushing: _____			
Running and retrieving tools: _____			
Test plug: _____			
Other requirements: _____			
FOOTNOTE			
^a If yes, specify what and by whom.			

Figure B.2—Wellhead Equipment Data Sheet—Casing-head Housing

Wellhead Equipment Data Sheet—Casing-head Spool			
Casing-head spool		PSL: _____	PR: _____
Bottom connector:		Size: _____	
		Rated working pressure: _____	
		Type: _____	
Top connector:		Size: _____	
		Rated working pressure: _____	
		Type: _____	
Outlets:		Size: _____	
		Rated working pressure: _____	
		Type: _____	
		Number: _____	
Equipment for outlets:		Valve-removal plug: _____	
		Valves (right): Qty _____	Typ _____ PSL _____ PR _____
		Valves (left): Qty _____	Typ _____ PSL _____ PR _____
		Companion flanges: Qty _____	Typ _____ PSL _____
		Bullplugs: Qty _____	
		Nipples: Qty _____	
		Needle valves: Qty _____	
		Gauges: Qty _____	
Lock screws? Yes _____	No _____	Lock screw function: _____	
Special material requirements: _____			
Bottom casing spool packoff size: _____			
		Type: _____	
		PR: _____	
Casing hanger:		Size: _____	
		Type: _____	
		PSL: _____	
		PR: _____	
Temperature rating (Table 2): _____			
Material class (Table 3): _____			
Retained fluid corrosivity (Table B.1): _____			
Witness? Yes ^a _____		No _____	
External coating? No _____ Yes _____		If yes, type _____	
Internal coating? No _____ Yes _____		If yes, type _____	
Flange bolting requirements (Table 16)		Exposed _____ Nonexposed _____	
Outlet inboard (studs): _____		(nuts): _____	
Outlet other (studs): _____		(nuts): _____	
Test and auxiliary equipment:			
Wear bushing: _____			
Running and retrieving tools: _____			
Test plug: _____			
Other requirements: _____			
FOOTNOTE			
^a If yes, specify what and by whom.			

Figure B.3—Wellhead Equipment Data Sheet—Casing-head Spool

Wellhead Equipment Data Sheet—Tubing-head Spool			
Tubing-head spool	PSL: _____	PR: _____	
Bottom connector:	Size: _____		
	Rated working pressure: _____		
	Type: _____		
Top connector:	Size: _____		
	Rated working pressure: _____		
	Type: _____		
Outlets:	Size: _____		
	Rated working pressure: _____		
	Type: _____		
	Number: _____		
Equipment for outlets:	Valve-removal plug: _____		
	Valves (right): Qty _____	Typ _____	PSL _____ PR _____
	Valves (left): Qty _____	Typ _____	PSL _____ PR _____
	Companion flanges: Qty _____	Typ _____	PSL _____
	Bullplugs: Qty _____		
	Nipples: Qty _____		
	Needle valves: Qty _____		
	Gauges: Qty _____		
Lock screws? Yes _____	No _____	Lock screw function: _____	
Material requirements: _____			
Bottom tubing spool packoff:	Size: _____		
	Type: _____		
	PR: _____		
Tubing hanger: Size: _____			
Type: _____			
PSL: _____			
PR: _____			
Back-pressure valve type: _____			
Surface-controlled subsurface valve control lines: _____			
Temperature rating (Table 2): _____			
Material class (Table 3): _____			
Retained fluid corrosivity (Table B.1): _____			
Witness? Yes ^a _____		No _____	
External coating? No _____	Yes _____	If yes, type _____	
Internal coating? No _____	Yes _____	If yes, type _____	
Flange bolting requirements (Table 16)		Nonexposed _____ Exposed _____ Exposed (low strength) _____	
Main run (studs): _____	(nuts): _____		
Outlet inboard (studs): _____	(nuts): _____		
Outlet other (studs): _____	(nuts): _____		
Test and auxiliary equipment:			
Wear bushing: _____			
Running and retrieving tools: _____			
Test plug: _____			
Other requirements: _____			
FOOTNOTE ^a If yes, specify what and by whom.			

Figure B.4—Wellhead Equipment Data Sheet—Tubing-head Spool

Wellhead Equipment Data Sheet—Crossover Flange	
Crossover flange	PSL: _____ PR: _____
Bottom connector:	Size: _____
	Rated working pressure: _____
	Type: _____
Top connector:	Size: _____
	Rated working pressure: _____
	Type: _____
Packoff type: _____	
Size: _____	
Temperature rating (Table 2): _____	
Material class (Table 3): _____	
Retained fluid corrosivity (Table B.1): _____	
Witness? Yes ^a _____ No _____	
External coating? No _____ Yes _____ If yes, type _____	
Internal coating? No _____ Yes _____ If yes, type _____	
Flange bolting requirement (Table 16) Nonexposed _____ Exposed _____ Exposed (low strength) _____	
Main run (studs): _____ (nuts): _____	
FOOTNOTE	
^a If yes, specify what and by whom.	

Figure B.5—Wellhead Equipment Data Sheet—Crossover Flange

Wellhead Equipment Data Sheet—Tubing-head Adapter	
Tubing-head adapter	PSL: _____ PR: _____
Bottom connector:	Size: _____
	Rated working pressure: _____
	Type: _____
Top connector:	Size: _____
	Rated working pressure: _____
	Type: _____
Surface-controlled subsurface safety valve outlets: _____	
Number: _____	
Size: _____	
Electrical feed-through connection? _____	
Special material requirements: _____	
Temperature rating (Table 2): _____	
Material class (Table 3): _____	
Retained fluid corrosivity (Table B.1): _____	
Witness? Yes ^a _____ No _____	
External coating? No _____ Yes _____ If yes, type _____	
Internal coating? No _____ Yes _____ If yes, type _____	
Flange bolting requirement (Table 16) Nonexposed _____ Exposed _____ Exposed (low strength) _____	
Main run (studs): _____ (nuts): _____	
FOOTNOTE	
^a If yes, specify what and by whom.	

Figure B.6—Wellhead Equipment Data Sheet—Tubing-head Adapter

Wellhead Equipment Data Sheet—Tree and Choke										
Tree	Single	Dual	Solid block	Stacked				External coating? If yes, state type	Flanged bolting requirements ^c Studs Nuts	Ring gasket type
			Size	Material ^a	PSL	PR	Witness? ^b			
Lower master valve: _____										
Upper master valve: _____										
Swab (crown) valve: _____										
Wing valve—inboard: _____										
Wing valve(s)—other: _____										
Tee/cross (circle one): _____										
Choke: _____										
End flange: _____										
Companion flanges: _____										
Instrument flanges: _____										
Tree cap/top conn.: _____										
Rated working pressure: _____										
Retained fluid corrosivity (Table B.1): _____										
Temperature rating (Table 2): _____										
Material class (Table 3): _____										
Upper master prepared for actuator:				Yes	____	No	____	If yes, specify Class I, II, or III below PR column		
Wing valve—inboard prepared for actuator:				Yes	____	No	____	If yes, specify Class I or II below PR column		
Wing valve—other prepared for actuator:				Yes	____	No	____	If yes, specify Class I or II below PR column		
Choke: adjustable or fixed: _____										
Orifice size: _____				Nominal size: _____						
Pressure drop: _____										
Flowline connector:		Size: _____								
		Type: _____								
Special material requirements: _____										
Other requirements: _____										
Upper master valve type actuator requirements:				Pneu./piston _____		Hydr./piston _____		Electric _____		
Supply pressure/power: _____				Pneu./diaphragm _____		Hydr./diaphragm _____		Electric _____		
Air _____ Gas _____										
Wing valve type actuator requirements:				Pneu./piston _____		Hydr./piston _____		Electric _____		
				Pneu./diaphragm _____		Hydr./diaphragm _____		Electric _____		
Supply pressure: _____										
Other: _____										

FOOTNOTES

^a Define or specify material requirements and, if cladding or other corrosion-resistant materials are to be inlaid, state base material type/clad material type, e.g. 4130/625.

^b If yes, specify what and by whom.

^c Indicate required bolting for the applicable retained fluid and temperature class specified in Table 17.

Wellhead Equipment Data Sheet—Multistage Crossover Spool	
Multistage crossover spool	PSL: _____ PR: _____
Bottom connector:	Size: _____
	Rated working pressure: _____
	Type: _____
Outlets lower:	Size: _____
	Rated working pressure: _____
	Type: _____
	Number: _____
Equipment for lower outlets:	Valve-removal plug: _____
	Valves (right): Qty: _____ Typ: _____ PSL: _____ PR: _____
	Valves (left): Qty: _____ Typ: _____ PSL: _____ PR: _____
	Companion flanges: Qty _____ PSL: _____
	Bullplugs: Qty _____
	Nipples: Qty _____
	Needle valves: Qty _____
	Gauges: Qty _____
Outlets upper:	Size: _____
	Rated working pressure: _____
	Type: _____
	Number: _____
Equipment for upper outlets:	Valve-removal plug: _____
	Valves (right): Qty: _____ Typ: _____ PSL: _____ PR: _____
	Valves (left): Qty: _____ Typ: _____ PSL: _____ PR: _____
	Companion flanges: Qty _____ PSL: _____
	Bullplugs: Qty _____
	Nipples: Qty _____
	Needle valves: Qty _____
	Gauges: Qty _____
Top connector:	Size: _____
	Rated working pressure: _____
	Type: _____
Special material requirements:	_____
Temperature rating (Table 2):	_____
Material class (Table 3):	_____
Retained fluid corrosivity (Table B.1):	_____
Lock screws? Yes _____ No _____	Lock screw function: _____

Figure B.8—Wellhead Equipment Data Sheet—Multistage Crossover Spool

Wellhead Equipment Data Sheet—Multistage Crossover Spool <i>(continued)</i>	
Casing hanger (lower):	Size: _____
	Type: _____
	PR: _____
	PSL: _____
Restricted-area packoff (lower):	Size: _____
	Type: _____
Temperature rating (Table 2):	_____
Material class (Table 3):	_____
Retained fluid corrosivity (Table B.1):	_____
Casing hanger (upper):	Size: _____
	Type: _____
	PR: _____
	PSL: _____
Restricted-area packoff (upper)	Size: _____
	Type: _____
Temperature rating (Table 2):	_____
Material class (Table 3):	_____
Retained fluid corrosivity (Table B.1):	_____
Flange bolting requirements (Table 16)	Nonexposed _____ Exposed _____ Exposed (low strength) _____
Outlet lower (studs): _____	(nuts): _____
Outlet upper (studs): _____	(nuts): _____
Main run (studs): _____	(nuts): _____
Other requirements: _____	
Witness? No _____ Yes ^a _____	
External coating? No _____ Yes _____	If yes, type: _____
Internal coating? No _____ Yes _____	If yes, type: _____
Test and auxiliary equipment: (top and/or bottom) _____	
Wear bushings: _____	
Running and retrieving tools: _____	
Test plugs: _____	
Other requirements: _____	
FOOTNOTE	
^a If yes, specify what and by whom.	

Figure B.8—Wellhead Equipment Data Sheet—Multistage Crossover Spool *(continued)*

Wellhead Equipment Data Sheet—Wellhead Safety Valves	
Wellhead safety valves	
General	
Special environmental conditions _____	Unusual ambient or operating temperatures, or atmospheric conditions conducive to corrosion or underwater use
Coating _____	
Shipping instructions _____	
SSV/USV valve	
(Class I, II, or III SSV/USV) _____	
Manufacturer _____	Model and type _____
Size _____	
Rated working pressure _____	
Temperature range _____	
SSV/USV actuator	
Manufacturer _____	Model and type _____
Cylinder rated working pressure _____	
Operating pressure _____	Purchaser to specify available supply pressure, if applicable
Temperature range _____	
Lock-open device _____	
USV _____	Working water depth _____

Figure B.9—Wellhead Equipment Data Sheet—Wellhead Safety Valves

Equipment Data Sheet—Boarding Shutdown Valves	
General	
Special environmental conditions _____	Unusual ambient or operating temperatures, or atmospheric conditions conducive to corrosion, or use in the offshore splash zone
Coating _____	
Shipping instructions _____	
BSDV valve	
(Class I, II, or III BSDV) _____	
Manufacturer _____	Model and type _____
Size _____	
Rated working pressure _____	
Temperature range _____	
BSDV actuator	
Manufacturer _____	Model and type _____
Cylinder rated working pressure _____	
Operating pressure _____	Purchaser to specify available supply pressure, if applicable
Temperature range _____	

Figure B.10—Equipment Data Sheet—Boarding Shutdown Valves

Wellhead Equipment Data Sheet—Choke Sizing						
Application						
Fluid						
Quantity						
End connectors/A&B Dimensions ^a						
Pressure rating/inlet			Outlet			
Temperature rating						
Material class		Body		Trim		
PSL		PR				
Service conditions at		Max. flow (units)	Normal flow (units)		Min. flow (units)	
Pressure	Inlet					
	Outlet or ΔP					
Temperature at inlet						
Oil	Flow rate					
	S.G. (if available)					
Gas	Flow rate					
	or G.O.R.					
	S.G. (if available)					
Liquid	Flow rate					
	S.G. (if available)					
Manual/actuated						
Actuator type/make/model						
Power source						
Manual override						
Position indication		Local		Remote/position transmitter		
Positioner						
Additional comments						
Adjustable or positive:						
Maximum orifice diameter:						
Type of flow bean:						
FOOTNOTE						
^a See Figure 16 and Figure 17.						

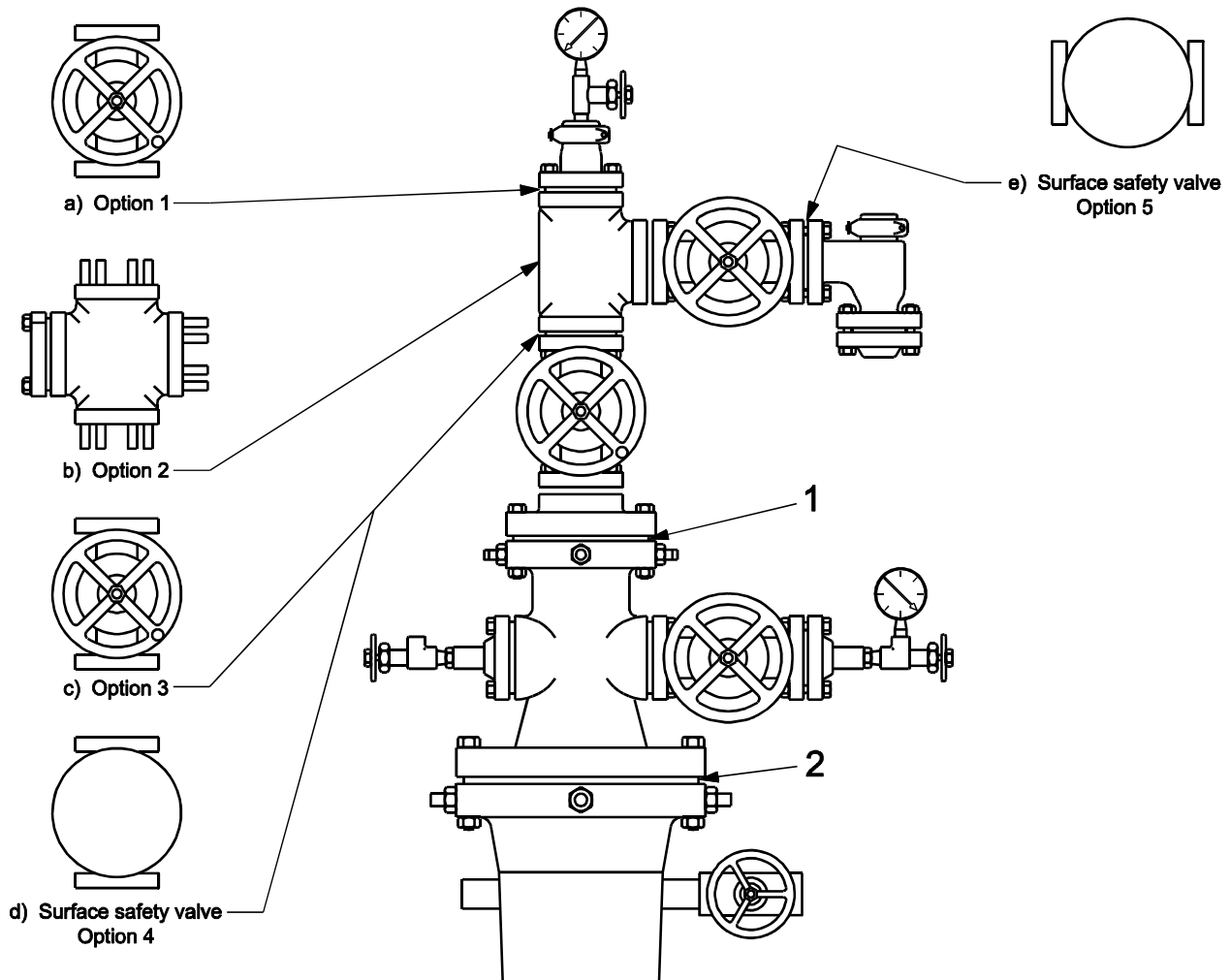
Figure B.11—Wellhead Equipment Data Sheet—Choke Sizing

Wellhead Equipment Data Sheet—Actuator and Bonnet					
Pneumatic Diaphragm Single _____ Double _____ Piston Single _____ Double _____	Hydraulic Conventional Rising stem _____ Nonrising stem _____ Wirecutter _____ Self-contained _____	Electric _____ _____ Wire/cable size _____ Stand-alone power source _____			
Supply requirements/specifications Pneumatic Availability _____ MPa (psi) Max. _____ Min. _____ Clean air _____ Nitrogen _____ Other _____		Hydraulic Availability _____ MPa (psi) Max. _____ Min. _____ Self-contained _____ Other _____			
Electric Voltage _____ DC _____ AC _____ Phase _____ Frequency _____ Current available _____ Other _____					
Actuator requirements/specifications Actuator Temperature rating (Table 2) _____ Materials class (Table 3) _____ External coating? No _____ Yes _____ If yes, type _____		Field data Customer _____ Field location _____ Platform _____ Well No. _____ Closed-in tubing-head pressure _____ MPa (psi) Accessories Fusible hold-open device _____ Manual hold-open device _____ Quick exhaust valve _____ Position indication a) local _____ b) remote _____			
Bonnet requirements Size _____ Model _____ Rated working pressure _____ MPa (psi) _____ Material class: _____ Temperature rating: _____					

Figure B.12—Wellhead Equipment Data Sheet—Actuator and Bonnet

B.3 Typical Wellhead and Tree Configurations

Examples of typical wellhead and tree configurations are shown in Figure B.13 and Figure B.14. Also included are examples of casing and bit programs that are consistent with the wellheads shown.

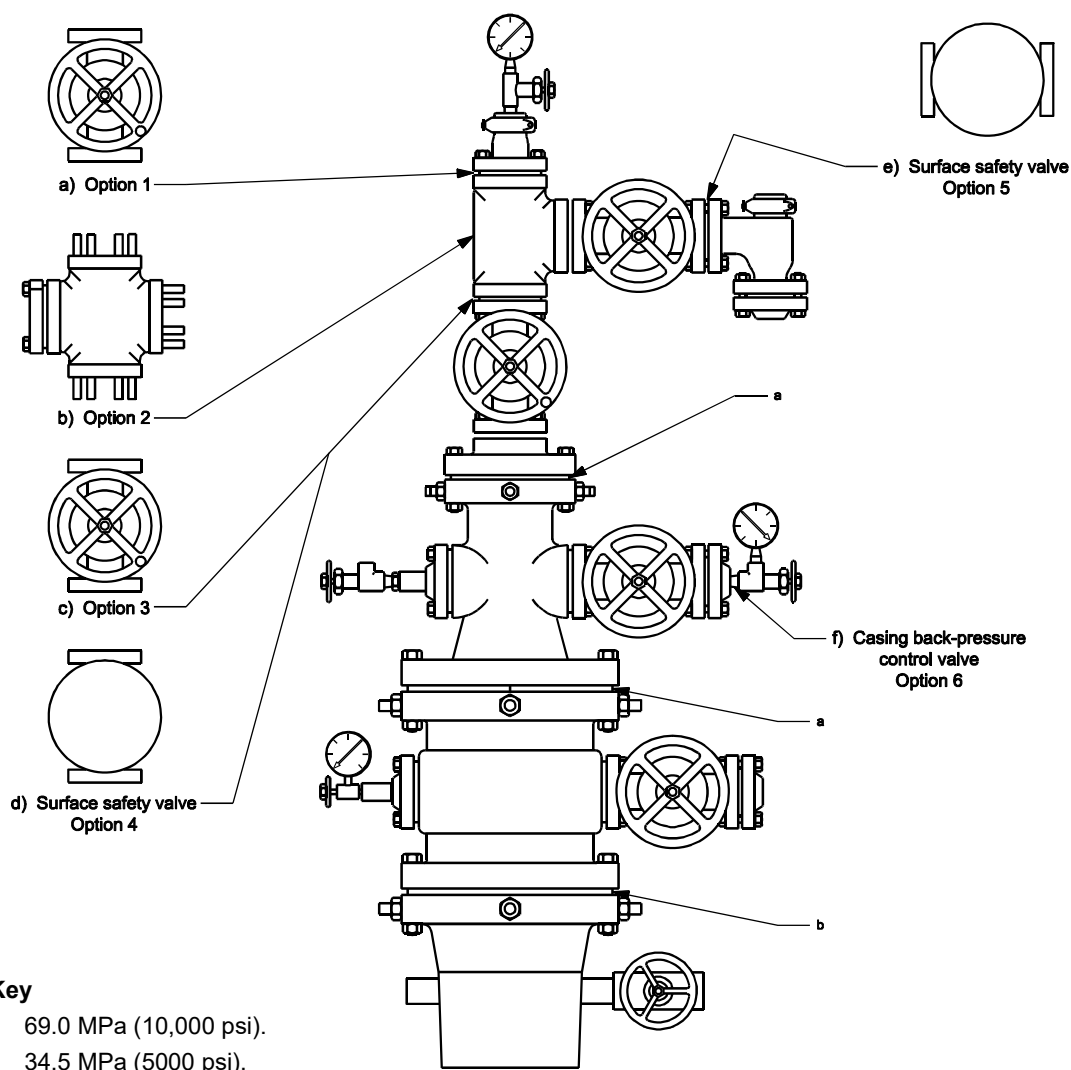


Key

- 1 tubing-head top flange 34.5 MPa (5000 psi)
- 2 casing-head top flange 20.7 MPa (3000 psi) or 34.5 MPa (5000 psi)

Typical Programs			
Casing Program mm (in.)	Bit Program mm (in.)	Casing-head Top Flange mm; MPa (in.; psi)	Tubing-head Top Flange mm; MPa (in.; psi)
219.1 (8 ⁵ / ₈) x 139.7 (5 ¹ / ₂)	200.0 (7 ⁷ / ₈)	279; 20.7 (11; 3000)	179; 34.5 (7 ¹ / ₁₆ ; 5000)
244.5 (9 ⁵ / ₈) x 177.8 (7)	215.9 (8 ¹ / ₂) or 222.2 (8 ³ / ₄)	or	
273.1 (10 ³ / ₄) x 193.7 (7 ⁵ / ₈)	250.8 (9 ⁷ / ₈)	279; 34.5 (11; 5000)	

Figure B.13—Typical Wellhead and Tree Configuration for a 34.5 MPa (5000 psi) Rated Working Pressure



Typical Programs				
Casing Program mm (in.)	Bit Program mm (in.)	Casing-head Housing Top Flange mm; MPa (in., psi)	Casing-head Spool Top Flange mm; MPa (in., psi)	Tubing-head Top Flange mm; MPa (in., psi)
406.4 x 273.1 x 193.7 (16 x 10 ³ / ₄ x 7 ⁵ / ₈)	374.7 x 250.8 or 241.3 (14 ³ / ₄ x 9 ⁷ / ₈ or 9 ¹ / ₂)	425; 34.5 (16 ³ / ₄ ; 5000)	279; 69.0 (11; 10,000)	179; 69.0 (7 ¹ / ₁₆ ; 10,000)
406.4 x 298.5 x 244.5 x 177.8 liner (16 x 11 ³ / ₄ x 9 ⁵ / ₈ x 7) liner	374.7 x 269.9 x 215.9 (14 ³ / ₄ x 10 ⁵ / ₈ x 8 ¹ / ₂)	425; 34.5 (16 ³ / ₄ ; 5000)	346; 69.0 (13 ⁵ / ₈ ; 10,000) 279; 69.0 (11; 10,000)	179; 69.0 (7 ¹ / ₁₆ ; 10,000)
339.7 x 244.5 x 177.8 (13 ³ / ₈ x 9 ⁵ / ₈ x 7)	311.2 x 215.9 x 152.4 (12 ¹ / ₄ x 8 ¹ / ₂ x 6)	346; 34.5 (13 ⁵ / ₈ ; 5000)	279; 69.0 (11; 10,000)	179; 69.0 (7 ¹ / ₁₆ ; 10,000)
273.1 x 193.7 x 127.0 (10 ³ / ₄ x 7 ⁵ / ₈ x 5)	250.8 x 165.1 (9 ⁷ / ₈ x 6 ¹ / ₂)	279; 34.5 (11; 5000)	279; 69.0 (11; 10,000)	179; 69.0 (7 ¹ / ₁₆ ; 10,000)

Figure B.14—Typical Wellhead and Tree Configuration for a 69.0 MPa (10,000 psi) Rated Working Pressure

B.4 Product Specification Levels

B.4.1 General

This specification establishes requirements for four PSLs. These four PSLs are independent of verification and validation of a product and define different levels of manufacturing requirements. Equipment with a PSL has specific quality control, material, and testing requirements for equipment covered by this specification (see 4.3.4).

PSL 1 represents the set of basic or minimum requirements for quality, material, and testing. All products to which a PSL applies minimally satisfy PSL 1 requirements for conformance with this specification. Higher level PSLs may be applied to the equipment by the manufacturer or may be specified by the purchaser, based on intended use of the equipment. SSVs, USVs, and BSDVs cannot be PSL 1.

PSL 2 contains all the quality and testing requirements of PSL 1 but imposes additional requirements as detailed in this specification. BSDVs cannot be PSL 2. The additional requirements include limitations to variance between the material QTC and the production material. It includes requirements for Charpy testing for service temperature -20°F and below, volumetric inspection of welds, and magnetic particle inspection of accessible well wetted surfaces.

PSL 3 contains all the quality and testing requirements of PSL 2 but imposes additional requirements as detailed in this specification. The additional requirements include restricted tolerance of material chemistry, changes to the material QTC in relation to the section thickness of the equipment parts. It includes Charpy testing for all service temperatures; volumetric inspection of all material in body, bonnets, flanges, and stems; wet magnetic particle inspection of all accessible surfaces; and extended hydrostatic test times.

NOTE The designation “PSL 3G” contains the same quality and testing requirements of PSL 3 with the addition of gas testing; end connectors and hangers do not require gas testing, so PSL 3G does not apply.

PSL 4 contains the highest level of quality and testing requirements for any product within this specification. PSL 4 equipment meets all the requirements of PSL 3 and would apply to high-pressure equipment used in sour service.

Figure B.15 shows the recommended specification level for primary equipment. Primary equipment of a wellhead assembly includes the following, as a minimum:

- tubing head;
- tubing hanger;
- tubing-head adapter;
- lower master valve.

All other wellhead parts are classified as secondary. The specification level for secondary equipment may be different than the level for primary equipment.

The selection of a PSL should be based on a quantitative risk analysis, which is a formal and systematic approach to identifying potentially hazardous events and estimating the likelihood and consequences to people, environment, and resources, of accidents developing from these events.

The following comments apply to the basic questions asked in Figure B.15.

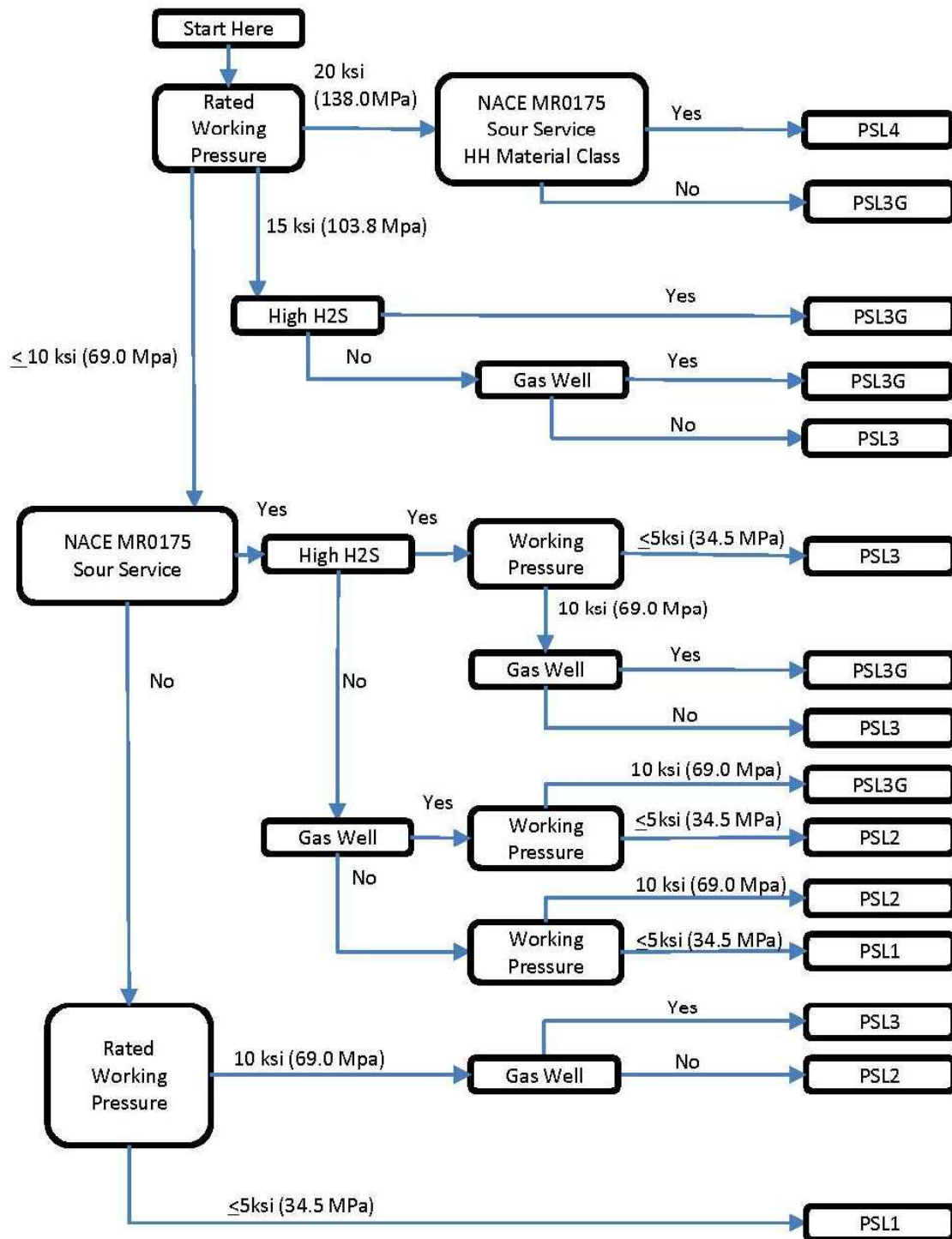


Figure B.15—Recommended Minimum PSL for Primary Parts